Measuring Cross-Lingual Robustness for Chemistry-Patent Retrieval: Patent-Grounded Multilingual Benchmarks, New Metrics, and a Deployment Decision

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Abstract

Chemical knowledge is often searched and reused across language boundaries: a user may ask a technical question in one language while the strongest supporting document is written in another. Retrieval systems should therefore be tested on whether they find that cross-language evidence, not only on whether they retrieve an easier same-language match. Yet multilingual chemistry retrieval has little clean public evaluation data, so models are often judged by a single aggregate recall score that can hide poor cross-language retrieval. We address this gap with public multilingual chemistry QAC benchmarks and a generation pipeline for cross-lingual retrieval evaluation. Google Patents and EPO releases provide controlled multilingual technical evidence, while chemistry ontologies enable chemically plausible hard negatives. We export the benchmarks in retrieval-ready form with corpus, queries, qrels, QAC records, and language metadata, enabling evaluation by query language, document language, same-language versus cross-language relevance, and chemical confusability. Aggregate Recall@10 overstates deployment readiness: the best cross-language Recall@10 is 0.50, same-language advantage reaches +0.55, and chemically confusable documents outrank gold evidence on 14–78% of publication-lens queries. The benchmark and pipeline provide a practical framework for building and auditing specialized multilingual retrieval evaluations in high-trust technical domains. Data and code are available on Hugging Face (Google Patents), Hugging Face (EPO), and GitHub.

1 Introduction

Technical search in chemistry rarely respects a single language boundary. A researcher may ask a question in English, German, French, Spanish, or another working language, while the strongest evidence may be documented elsewhere: in a translated patent family, in a regional filing, or in technical text whose terminology follows a different linguistic convention. This is not only a translation problem. Chemistry search depends on aliases, abbreviations, formulae, compound families, process descriptions, and near-neighbor concepts that can be semantically close while still being wrong for the user's question. For a retrieval system used in a high-trust technical workflow, returning a plausible but chemically different document is not a harmless ranking error.

This matters in multinational industrial settings, where technical documents may be stored in several languages but downstream users and systems need the relevant evidence regardless of the query language or document language. The same retrieval requirement appears in RAG systems, agentic workflows, and other decision-support pipelines that depend on finding the right evidence before reasoning over it. Translating every query or document can help in some cases, but it becomes less attractive as language coverage grows because translation adds latency, cost, and another source of technical-term errors.

Dense retrieval with neural sentence and passage embeddings (Reimers and Gurevych, 2019; Karpukhin et al., 2020) is attractive for this setting because multilingual embedding models promise a single retrieval index over many languages (Feng et al., 2022; Chen et al., 2024). In practice, however, the evaluation signal used to choose such a model is often too coarse. A model can achieve a strong average Recall@k by retrieving easy same-language evidence, while failing when the intended evidence is in another language, a bias also observed in recent cross-lingual retrieval analyses (Goworek et al., 2025; Amiraz et al., 2025). The problem is not only whether a multilingual encoder retrieves a relevant document, but whether it retrieves the right document when query language, document language, and chemical similarity pull the ranking in different directions. This is the failure mode that matters in deployment, and it is precisely the one that an aggregate retrieval score can hide.

Existing multilingual retrieval benchmarks provide important coverage for general cross-lingual search (Zhang et al., 2023; Lawrie et al., 2023; Muennighoff et al., 2023; Enevoldsen et al., 2025), but they do not directly capture this chemistry workflow. We need evaluation data where language variants are controlled, questions are grounded in technical contexts, and hard negatives are chemically plausible rather than arbitrary. Without this structure, an evaluation may reward the model that looks best on average while missing the model behavior that creates practical risk.

We present a multilingual chemistry question-answer-context (QAC) benchmark and generation pipeline for cross-lingual retrieval evaluation. Our goal is to evaluate multilingual chemistry retrieval, not patent search alone. We use patents as the benchmark source because patent families often contain related technical documents in multiple languages and include chemistry-rich descriptions at scale, building on the long history of patent retrieval evaluation (Piroi et al., 2011).

This paper is therefore an evaluation-framework paper rather than a model leaderboard. Our contributions are: (1) to our knowledge, the first public multilingual chemistry retrieval benchmark with query/document language metadata and retrieval-ready corpus, queries, and qrels; (2) a reproducible LLM-assisted QAC generation and validation pipeline; (3) a language-aware retrieval evaluation that separates same-language and cross-language evidence; (4) a chemistry-confusability stress test based on alias and ontology neighbors; and (5) a deployment-oriented model-selection analysis showing why aggregate recall is not a sufficient deployment dashboard.

2 Related Work

General embedding benchmarks such as MTEB and MMTEB, and multilingual or cross-lingual retrieval resources such as MIRACL, NeuCLIR, and CLIRMatrix, have made retrieval and embedding evaluation easier to compare across models and languages (Zhang et al., 2023; Lawrie et al., 2023; Muennighoff et al., 2023; Enevoldsen et al., 2025; Sun and Duh, 2020). They frame our evaluation setting, but they are not designed around chemistry-specific relevance, controlled language variants, or chemically plausible hard negatives. Our benchmark follows this line of work while adding domain structure needed to audit multilingual retrieval in technical chemistry search.

Chemistry-specific benchmarks have recently expanded evaluation beyond generic NLP tasks. ChemTEB evaluates chemical text embeddings across retrieval, classification, clustering, bitext mining, and pair-classification tasks (Kasmaee et al., 2024); ChemLit-QA provides expert-validated QAC data for chemistry RAG (Wellawatte et al., 2024); and ChemComp and ChemKGMultiHopQA test chemistry-focused multi-hop scientific reasoning in retrieval-aware settings (Khodadad et al., 2026; Astaraki et al., 2026). Recent chemical literature embedding work such as ChEmbed further shows the value of domain-adapted retrieval models and chemistry-specific retrieval evaluation (Kasmaee et al., 2025). Our work is complementary: it focuses on multilingual retrieval robustness, controlled language variants, and chemistry-confusable hard negatives rather than chemistry QA, reasoning, or embedding evaluation alone.

Recent work on cross-lingual retrieval and multilingual RAG shows that model performance and behavior can depend strongly on language, alignment, and same-language preference (Goworek et al., 2025; Amiraz et al., 2025; Li et al., 2024; Liu et al., 2025; Ki et al., 2025). These findings motivate our focus on separating same-language from cross-language relevance instead of relying only on aggregate recall. In contrast to settings where content and language are difficult to disentangle, patent publication variants give us a practical way to compare multilingual technical evidence under tighter content control.

Patent retrieval has a long evaluation history, including NTCIR and CLEF-IP (Fujii and Ishikawa, 2002; Piroi et al., 2011), and recent work has revisited patent embeddings (Ghosh et al., 2024; Yousefiramandi and Cooney, 2026). Our goal is complementary: we do not propose a new encoder, but a chemistry-grounded evaluation framework. We use chemical ontology and alias structure, including ChEBI (Hastings et al., 2016), to define hard negatives that are plausible for chemistry search but wrong for the intended evidence, complementing ontology-grounded extraction work such as CEAR (Langer et al., 2024).

3 QAC Generation Pipeline

Our method turns multilingual chemistry technical text into auditable question-answer-context (QAC) records for retrieval evaluation. The design goal is not simply to generate fluent questions. Each generated item must identify the evidence document it came from, preserve language metadata, and carry quality signals that make the benchmark inspectable after export. Figure 1 summarizes the full pipeline.

Corpus and context construction. The input is a chemistry-relevant multilingual corpus. In this work, we use two patent-derived sources, Google Patents and EPO, because patents provide technical descriptions and multilingual publication variants at scale. Both sources are normalized into the same document-language representation before QAC generation. For each document-language pair, the context builder assembles a compact evidence block from the title and abstract, adding first-claim text when available; longer description fields are source-dependent and are not assumed to be present for every document. This step keeps the generation grounded in a specific document rather than asking the model to produce generic chemistry questions from memory.

Candidate generation. For each selected document, language, and generation mode, an LLM produces multiple QAC candidates. Each candidate contains a user-facing question, a short answer, the supporting context, and metadata linking the item back to the source document and language. We use both technical and semantic question modes so that the benchmark contains queries that reflect different retrieval intents: some ask for concrete technical details, while others ask for a higher-level description of what the document is about.

Verification and filtering. Generated candidates are then passed through an LLM-based verifier. The verifier scores whether the answer is faithful to the provided context, whether the question is answerable from that context, whether the language is appropriate, and whether the item is useful as a retrieval query. We retain the best-scored items together with their audit metadata rather than treating generation as a black box. This gives each QAC row a provenance trail: source document, language, generation mode, scores, and verifier feedback.

Human audit and export. Because the verifier is itself model-based, we audit its behavior after generation against a human-annotated sample. This does not make human validation an automatic step for every item, but it checks whether the automatic grading is directionally reliable enough for benchmark construction. The filtered QAC records are then exported with the retrieval artifacts used in later sections: a corpus table, query table, relevance judgments, and the full QAC table. These artifacts let us evaluate models with standard dense-retrieval tooling while still slicing results by query language, document language, and source provenance.

4 Benchmark Construction

The filtered QAC records are exported as retrieval benchmarks rather than as a single flat table. We release two patent-derived benchmark sources: one based on Google Patents and one based on EPO data. Each release contains four linked views: `corpus` stores the searchable documents and their languages; `queries` stores the generated questions; `qrels` stores relevance judgments from queries to documents; and `qac` preserves the original question, answer, context, source document, generation metadata, and quality scores. This separation lets standard retrieval evaluators consume the data while keeping the provenance needed to audit failures by language and source. Table 1 summarizes the released dataset statistics.

Cross-language relevance. For the multilingual chemistry retrieval benchmark, each query is tied to a source technical document. Relevant documents include the source document and its available multilingual variants, so a query can be evaluated against both same-language and cross-language evidence. We keep query language, document language, and source identifiers as explicit metadata. This makes it possible to separate an easy same-language hit from a harder cross-language hit, and to construct "no-home" cases in which the query language has no same-language gold document. These cases are central to the benchmark because they remove the shortcut that often inflates aggregate recall.

Chemistry-aware hard negatives. Language is only one source of retrieval error in chemistry. A model may also rank a document about a related compound, synonym, parent class, or neighbor-

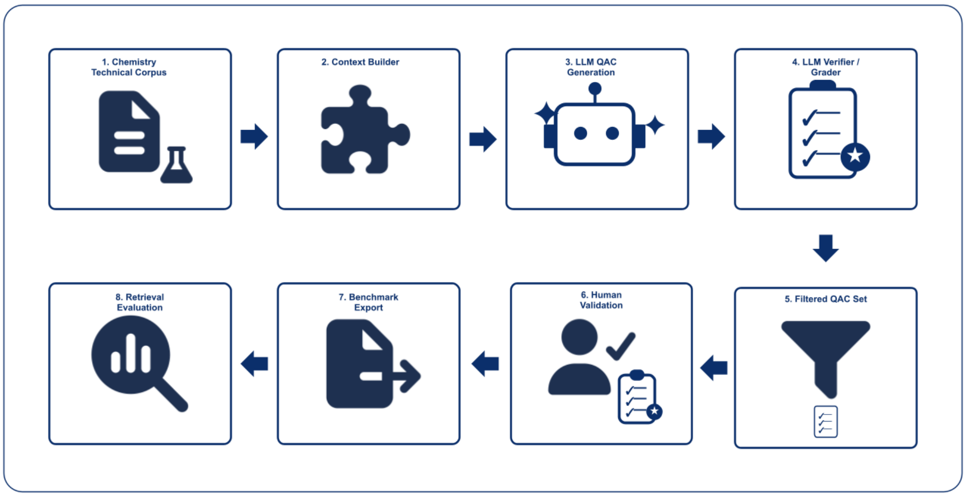

Figure 1: Overview of the multilingual chemistry retrieval benchmark pipeline, from chemistry technical text through QAC generation, verification, human audit, benchmark export, and retrieval evaluation.
Figure note: Icons adapted from Font Awesome Free, licensed under CC BY 4.0.

Source	Corpus	Queries	Qrels	Cross-lang qrels	Q tokens	Q vocab	C tokens	C vocab
Google Patents	23,787	524	1,284	1,023	11.3 ± 6.3	2,742	145.0 ± 63.8	83,333
EPO	11,315	198	594	396	14.1 ± 5.4	1,371	178.2 ± 71.1	72,256

Table 1: Statistics for the two released patent-derived multilingual QAC retrieval benchmarks. Question and corpus token columns report mean $\pm$ standard deviation using regex token counts. Vocabularies are unique case-folded tokens.

ing concept above the intended evidence. To test this failure mode, we build an alias-graph stress test from chemistry ontology and alias structure. ChEBI concepts and multilingual aliases define target chemical concepts; documents associated with the target concept form the gold set; and taxonomic or alias-graph neighbors provide chemically plausible hard negatives. These negatives are deliberately more informative than random non-relevant documents: they ask whether the retriever can distinguish the right chemical evidence from evidence that is close enough to be tempting.

The resulting benchmark labels each retrieved document by its role in the workflow: same-language relevant evidence, cross-language relevant evidence, unrelated distraction, or chemically confusable hard negative. The next section uses these labels to evaluate multilingual embedding models beyond aggregate recall.

5 Evaluation Setup

We evaluate multilingual embedding models as drop-in dense retrievers. Each model embeds the query and all corpus documents; documents are ranked by embedding similarity; and retrieval quality is computed from the benchmark `qrels`. This setup matches the practical decision faced by a technical search team: choose one multilingual encoder for a shared index, then ask whether it retrieves the right evidence across languages.

Models and protocol. We evaluate nine multilingual or domain-relevant embedding models: `embeddinggemma`, `bge-m3` (Chen et al., 2024), `granite-278m`, `nomic-v2-moe`, `qwen3-0.6B`, `LaBSE` (Feng et al., 2022), `SapBERT` (Liu et al., 2021), `e5-large-instruct`, and `gte-base`. All models retrieve against the same multilingual chemistry corpus, and all queries are evaluated with the same relevance judgments. We use standard Recall@k as the baseline retrieval metric because it is familiar and easy to compare, but we treat it as only the starting point: average recall does not say where the relevant document came from or which failure mode produced a miss.

Language-aware views. For each model, we split results by query language, document language, and whether a retrieved relevant document is in the same language as the query or in an-

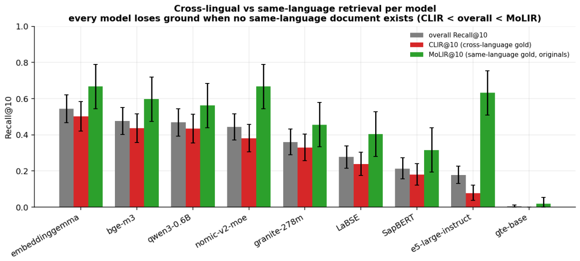

Figure 2: Evaluation separates aggregate Recall@10 from same-language retrieval and cross-language retrieval. The split is the central diagnostic: a model can look strong on average while still losing substantial ground when the relevant evidence is in another language.

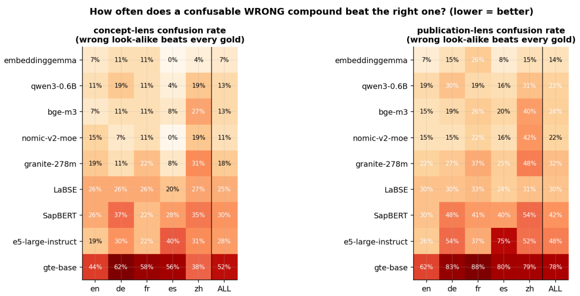

Figure 3: The alias-graph stress test measures chemistry-confusability: cases where a chemically plausible but incorrect document outranks all gold evidence. This complements language-aware recall by testing whether models distinguish the intended chemical concept from near neighbors.

other language. This gives two central views: same-language retrieval, where a model can often exploit lexical or embedding alignment advantages, and cross-language retrieval, where the model must bridge the language boundary. We also inspect language-pair behavior as a directed matrix, since retrieving a German document from an English query need not be as easy as retrieving an English document from a German query.

Failure analysis. Beyond relevant-document recall, we measure the kinds of mistakes that matter for the workflow. First, we track same-language distractions: cases where a model retrieves a non-relevant document in the query language ahead of relevant cross-language evidence. Second, we use the alias-graph benchmark to measure chemistry confusion, where a document about a chemically related but incorrect concept outranks the intended evidence. These analyses turn retrieval output into diagnostic categories rather than a single score.

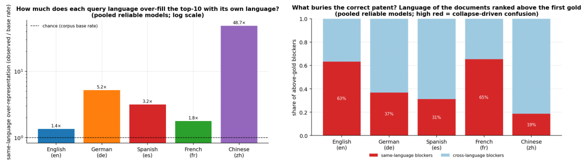

Figure 4: Language-collapse diagnostics for the cross-lingual benchmark. Same-language documents are over-represented in retrieved lists, and same-language distractors can bury cross-language gold evidence. This explains why average recall can hide the failure most relevant to multilingual chemistry search.

6 Results

We organize the results around the three questions the benchmark is meant to answer: whether generated QACs are usable, whether cross-language retrieval is harder than same-language retrieval, and whether chemistry-aware hard negatives surface failures that ordinary relevance metrics hide.

QAC quality is high enough for benchmark construction. Human review of 97 generated QACs gives a mean score of 8.33/10, with 94 items falling in the "good" bucket. Technical questions score slightly higher than semantic questions on average (8.72 vs. 7.91), but both modes remain usable for retrieval evaluation. The LLM verifier is slightly stricter than the human annotator (79.0% vs. 83.3%), supporting its use for filtering and auditing while still requiring human validation for benchmark-quality claims.

Average recall hides cross-language weakness. Figure 2 shows that same-language retrieval is consistently easier than cross-language retrieval: the best cross-language Recall@10 is 0.50, and same-language advantage reaches +0.55. A model can therefore look acceptable under an aggregate score while failing on the deployment-relevant case where the evidence is available only in another language. The directed language-pair view further shows that this difficulty is not symmetric: query language and document language both matter.

The errors are not only multilingual; they are chemical. The alias-graph benchmark shows a second failure mode. Even when a retrieved document is topically close, it may be chemically wrong. Figure 3 measures how often a chemically confusable document outranks all gold evidence; on the publication lens, this occurs on 14–78% of queries depending on the model. These failures are especially important for chemistry search because a nearby compound, parent class, or sibling concept can look plausible while answering the wrong question.

Together, the results support the central lesson of the benchmark: multilingual chemistry retrieval should be evaluated with language position and chemical confusability visible, not only with a single averaged retrieval score.

7 Deployment Decision

The evaluation is meant to support a practical model-selection decision: choose one multilingual embedding model for a shared chemistry retrieval index. If the team looked only at aggregate retrieval scores, the decision would be hard to audit because same-language hits and chemically plausible distractors are mixed into one number. Our benchmark changes the decision criterion from "which model has the best average Recall@10?" to "which model remains usable when the evidence is cross-language and chemically confusable?"

Capability vs. reading cost. Figure 5 shows the resulting cost-vs-capability view. We use cross-language Recall@10 as the capability axis and XRC50, the median cross-lingual reading-cost multiplier, as the cost axis. The Pareto frontier has three models: `embeddinggemma`, `bge-m3`, and `granite-278m`. `embeddinggemma` is the unique maximum-CLIR corner (CLIR@10 $=$ 0.5024, XRC50 $=$ 3.5), so it is the recommended capability-first choice. If an operating threshold of CLIR@10 $\geq$ 0.40 is acceptable, `bge-m3` becomes the cheaper-to-read admitted model (XRC50 $=$ 2.0). At a lower threshold, `granite-278m` enters as an even cheaper frontier point (XRC50 $=$ 1.25, CLIR@10 $=$ 0.3285). Thus the benchmark does not produce a context-free winner; it exposes the trade-off a deployment team must choose.

Decision rule. For a deployment that prioritizes maximum cross-language recall, we would select `embeddinggemma`. For a deployment with a stricter reading-cost budget and an acceptable CLIR@10 threshold around 0.40, `bge-m3` is the more efficient alternative. In both cases, the decision uses language-aware and chemistry-aware diagnostics rather than aggregate recall alone. This is the practical lesson: model selection for high-trust multilingual chemistry search should report capability, language robustness, and chemistry-confusability side by side before a retriever is shipped.

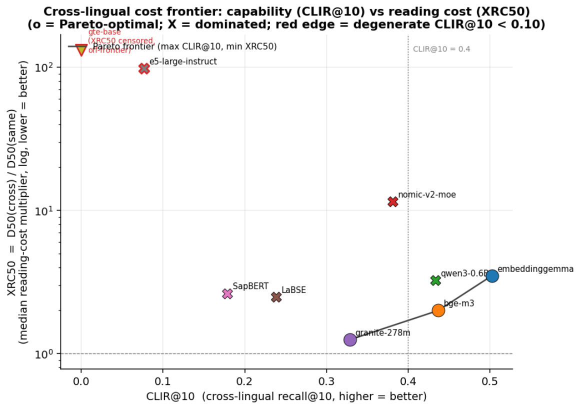

Figure 5: Deployment-oriented cost-vs-capability frontier. Cross-language Recall@10 measures capability, while XRC50 measures how much deeper a user must read to reach cross-language evidence. `embeddinggemma` is the maximum-capability corner; `bge-m3` is the cheaper-to-read option at a CLIR@10 threshold of 0.40.

Limitations

This benchmark is intentionally specialized. Chemistry patents provide a useful substrate for controlled multilingual technical evidence, but results may not transfer directly to other scientific domains, regulatory corpora, or general web retrieval. Patent publication variants also give stronger content control than unrelated multilingual corpora, but they do not guarantee full claim-level equivalence across every language version. The design should therefore be read as a reusable evaluation pattern, not as proof that the same model ordering will hold everywhere or that all patent variants are legally interchangeable.

The QAC pipeline relies on LLM generation and LLM-based verification. Human review supports the usefulness of the generated data, but the validation sample is limited and does not replace expert patent or chemistry review for every item. Future versions should expand expert annotation, especially for borderline chemical relevance and hard-negative severity.

Finally, the current paper focuses on multilingual retrieval and alias-graph confusability. Code-switching and noisy-query infrastructure exist in the project, but we do not claim complete results for those settings here.

8 Conclusion

Multilingual chemistry retrieval needs evaluation data that makes the right failure modes visible. In this paper, we presented a QAC generation and validation pipeline for building such data, released it in retrieval-ready form, and used it to evaluate dense multilingual retrievers beyond aggregate recall. Language metadata separates same-language from cross-language relevance; QAC provenance makes generated queries auditable; human validation checks the LLM-based verifier; and alias-graph hard negatives expose chemistry-specific confusions.

The central lesson is simple: a model that looks effective on average may still prefer same-language shortcuts or chemically plausible but wrong documents over the intended cross-language evidence. For high-trust multilingual retrieval, the question is not only whether the system retrieves something relevant on average, but whether it finds the intended evidence when language and chemical similarity create tempting shortcuts.

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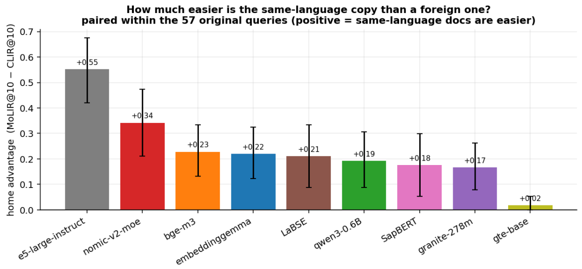

Figure 6: Same-language home advantage by model. The gap between same-language and cross-language retrieval is the main reason aggregate recall can overstate deployment readiness.

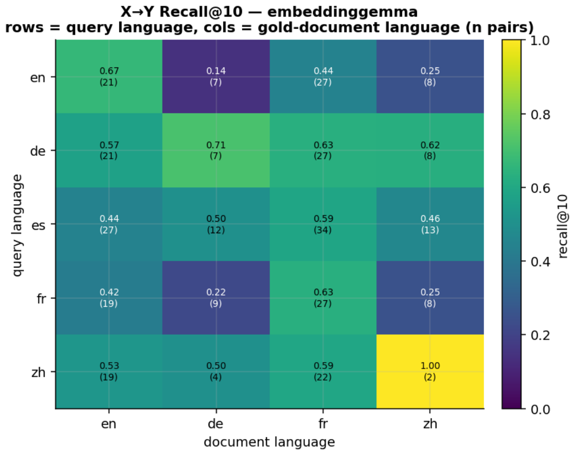

Figure 7: Directional query-language → document-language retrieval. Cross-language retrieval is not symmetric: a model's behavior depends on both the query language and the target document language.

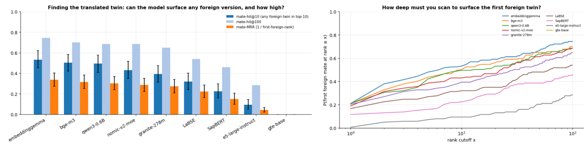

Figure 8: Foreign-twin retrieval diagnostics. The figure reports whether multilingual publication variants are found and how deeply a model must search before reaching the first cross-language twin.

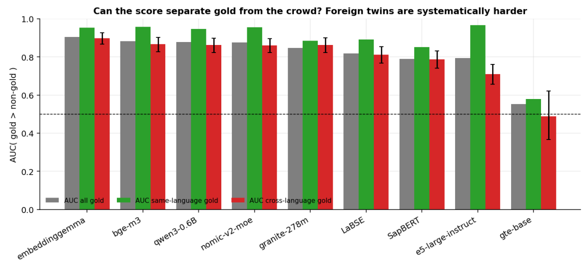

Figure 9: Embedding separability for same-language and cross-language gold documents. Lower cross-language separability indicates that the retrieval failure is partly representation-level, not only a top-k ranking artifact.

A Appendix

This appendix collects supporting analyses that are useful for interpretation but too detailed for the six-page body.

B Additional Cross-Lingual Retrieval Diagnostics

These diagnostics expand the paper's main cross-language retrieval result. They show that the same-language advantage is not a single-score artifact: it appears as model-level home advantage, directed language-pair asymmetry, delayed recovery of multilingual publication variants, and weaker embedding separability for cross-language gold evidence.

C Additional Alias-Graph Diagnostics

The alias-graph analyses isolate the chemistry side of the retrieval problem. They test whether multilingual queries for the same concept produce consistent rankings and whether chemically plausible neighbors become systematic distractors. Together, these figures explain why relevance in chemistry search cannot be reduced to topical similarity alone.

D Deployment-Oriented Supporting Analyses

These figures support the deployment interpretation in the main paper. They are secondary to the main cost-vs-capability decision, but they show two practical constraints: query translation can change retrieval behavior, and re-rankers can only recover cross-language evidence that appears within a feasible candidate pool.

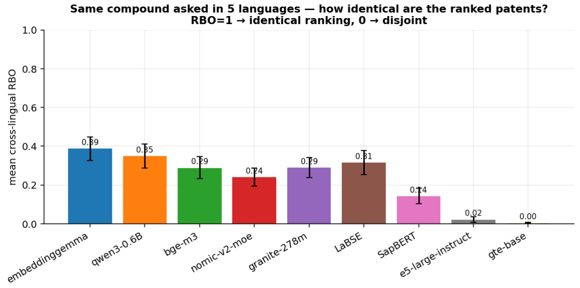

Figure 10: Cross-lingual ranking consistency for the same chemical concept asked in multiple languages. Low overlap means language variants of the same query can return substantially different ranked lists.

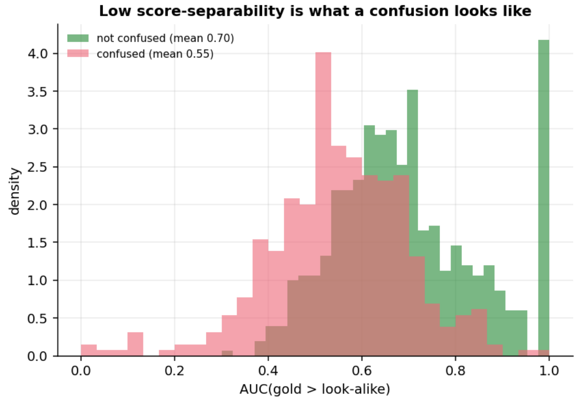

Figure 13: Confusion as separability collapse. Confused examples have weaker embedding separation between gold and non-gold documents.

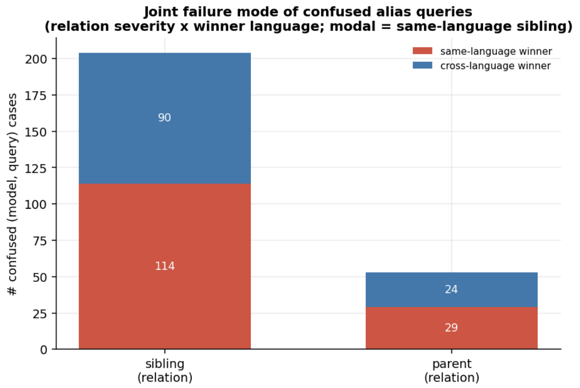

Figure 11: Joint language and chemistry failure modes among confused cases. This analysis links the paper's two central risks: same-language preference and chemically plausible but wrong neighbors.

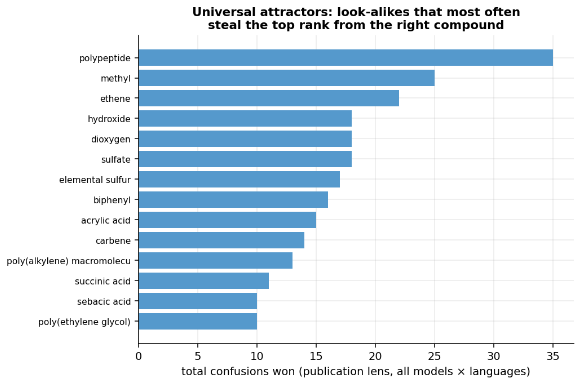

Figure 14: Universal attractors in the alias graph: chemically plausible look-alikes that repeatedly steal top rank across models or languages.

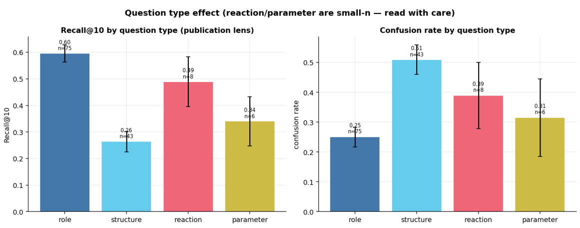

Figure 12: Question-type effects in the alias-graph benchmark. Some query types are substantially more vulnerable to chemistry-confusability than others.

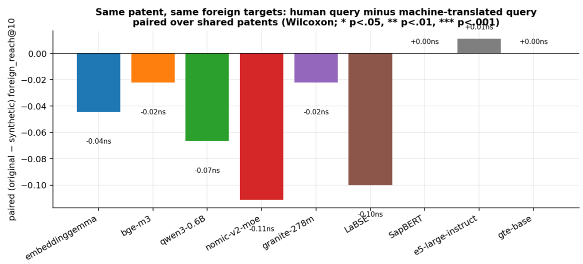

Figure 15: Human-written versus machine-translated queries. This supports the paper's practical separation between generating query variants and relying on human-translated source evidence.

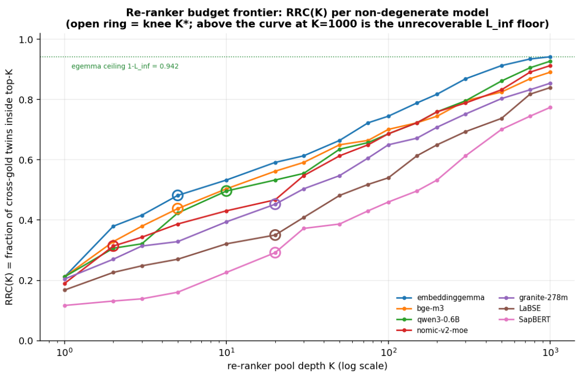

Figure 16: Re-ranker recoverability curves. These curves show how much cross-language evidence is recoverable at different candidate-pool depths and why some failures must be fixed before re-ranking.